🎬 MENAHIL FATIMA

A MENAHIL FATIMA ORIGINAL

Computer Science · Cybersecurity · AI/ML · Full-Stack Development · Mobile Apps
Beaconhouse National University · CGPA 3.79 · Dean's Honor List · Third Year

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WHO'S WATCHING?

💼 RECRUITER	🎓 STUDENT	🌐 EXPLORER
Hiring Manager Mode	Peer & Collaborator Mode	Just Browsing? Welcome.

▸ ABOUT THIS SERIES

Menahil Fatima is a third-year Computer Science student at Beaconhouse National University, Lahore — building at the intersection of cybersecurity, artificial intelligence, and full-stack development. This portfolio is a complete, season-by-season record of every project, internship, late-night debugging session, award, and lesson learned from Year 1 to present.

Selected for Pakistan's inaugural APAC Cyber Clinic cohort, co-designed with Google.org, The Asia Foundation, and the Global Cyber Alliance. Recipient of the AI Innovation Distinction Award from Devsinc CEO Usman Asif. IEEE-published researcher. Dean's Honor List every semester without exception. Merit Scholarship holder. This is not just a portfolio — it is a production.

- **GENRES:** Cybersecurity · AI & ML · Full-Stack Web · Mobile · Digital Forensics · NLP · Database Systems · Social Impact Tech
- **STATUS:** 🟢 Ongoing — Season 6 (Spring 2026) currently in production
- **RATED:** CS — Suitable for hiring managers, technical peers, professors, and curious humans

- **LANGUAGES:** Python · C++ · JavaScript · SQL · Dart · English · Urdu
- **HOME BASE:** Lahore, Pakistan
- **AVAILABLE:** Internships — Cybersecurity · Software Engineering · AI/ML Research · Full-Stack
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📺 SEASON 1 — THE ORIGIN STORY

Fall 2023

Year one. Two web projects, a steep HTML/CSS learning curve, and the first glimpse of what building things actually feels like. Every great developer has a Season 1. This is mine.

EPISODE 01 — Arcelia — Luxury Hotel Website

92% Match

Where elegance meets the DOM. The very first deployment.

- **Period:** Nov 2023 – Dec 2023
- **Course:** Website Development — Semester 1
- **Genre:** Front-End · Hospitality Tech · UI/UX · Responsive Design
- **Status:** ✅ Completed

THE PLOT

Arcelia is the first real website ever built — a fully designed, multi-page hotel experience for a fictional five-star property on Sentosa Island, Singapore. Every layout decision, every colour choice, every hover effect was deliberate. This project was the introduction to thinking like a developer AND a designer simultaneously: how does a page make someone feel? Does it load correctly on mobile? Are the buttons where a human would expect them? It was the project that answered the question: can I actually build things? The answer was yes.

KEY FEATURES

- Sign Up / Authentication — user registration flow with form validation
- Restaurant Showcase — full dining experience page with rich visual hierarchy
- Accommodation Listings — room categories with descriptive content and imagery layout

- Sustainability Section — environmental commitment and initiatives page
- Contact Us — functional contact form with field validation
- Responsive Design — cross-device layout compatibility throughout
- Navigation — sticky header with smooth section-to-section linking

TECH STACK: HTML5 · CSS3 · Front-End Development

👥 **COLLABORATORS:** Manahil Tanweer, Hania Atta

💡 **WHAT I LEARNED:** Design decisions ARE technical decisions. Padding, line-height, and colour contrast are not aesthetic preferences — they are user experience engineering. Learning this in Semester 1 changed how every subsequent project was approached.

EPISODE 02 — Dandelions — Botanical Garden Website

100% Match 🏆🏆

The one that won twice. The project that made it clear this was the right path.

- **Period:** Nov 2023 – Jan 2024
- **Course:** Website Development — Semester 1
- **Genre:** Front-End · E-Commerce · Award Winner · Educational Platform
- **Status:** ✅ Completed · Double Award Winner
- 🏆 Best Team Project — Web Development (NETSOL × Ayub Ghauri, 2024)
- 🏆 Best Team Project — Programming (NETSOL × Ayub Ghauri, 2024)

THE PLOT

What started as a semester deliverable became the first award-winning project. Dandelions is a full-featured botanical garden website for a fictional garden in Singapore — built to educate, engage, and inspire. The project was recognized at the 2024 BNU Project Showcase across two separate categories simultaneously, a rare double sweep. Beyond the technical achievement, Dandelions was the creative turning point — the moment a project stopped feeling like coursework and started feeling like something worth being proud of. The e-shop, learning modules, and gallery were each designed with intention, not just instruction.

KEY FEATURES

- Sign In / Register — user authentication with session handling and persistent login state
- E-Shop — plant catalogue with product pages, cart functionality, and checkout flow
- Activities & Events — listing and filtering of garden events and workshops
- Sustainability Efforts — environmental initiatives dedicated section
- Interactive Gallery — visual showcase of botanical species and garden spaces

- Learning Through Plants — educational module with plant profiles and interactive facts
- Contact Us — inquiry form with embedded location
- Responsive Design — Bootstrap grid system for seamless mobile and desktop experience
- Cross-Device Compatibility — tested across screen sizes and browsers

TECH STACK: HTML5 · CSS3 · Bootstrap 5 · JavaScript (Vanilla) · Front-End Development

👥 **COLLABORATORS:** Manahil Tanweer, Hania Atta

💡 **WHAT I LEARNED:** Winning isn't the goal — building something genuinely useful is. The awards followed naturally from caring about the user experience at every step. That lesson holds for every project since.

📺 SEASON 2 — THE C++ ARC

Spring 2024

The semester that introduced proper computer science. Algorithms, OOP, and the satisfying brutality of a terminal-based application with no framework to hide behind. One project — but a foundational one.

EPISODE 01 — Gaming Hub — Terminal Arcade Engine

95% Match

A terminal-based arcade built on pure C++ and raw logic. No shortcuts. No GUI. Just code.

- **Period:** Apr 2024 – Jun 2024
- **Course:** Programming Fundamentals — Semester 2
- **Genre:** Game Development · C++ · OOP · Data Structures & Algorithms · Retro Terminal
- **Status:** ✅ Completed

THE PLOT

Gaming Hub is a console application that hosts five distinct single-player games inside one unified platform — built entirely in C++. No game engines, no libraries, no frameworks. Just

Object-Oriented Programming, custom data structures, and a terminal. The challenge was not just building games, but architecting a system that manages user accounts, persists scores across sessions, scales difficulty, rewards milestones, and maintains state — all within a command-line interface. This project introduced thinking architecturally: how do you organize code so that adding a sixth game doesn't break the other five? These questions shaped every design decision — and their answers became the foundation for every system built afterward.

KEY FEATURES

- Login & Sign Up — user account creation with encrypted storage and persistent sessions
- Player Profiles — personalized dashboards with scores, progress history, and statistics per game
- Five Distinct Games — each with independent logic, win/loss conditions, and scoring mechanics
- Difficulty Scaling — Easy / Medium / Hard modes with adaptive challenge parameters per game
- Achievements System — milestone-based badge rewards (e.g., 'Win 10 in a row', 'Reach 1000 points')
- Leaderboards — score ranking with historical comparison and self-competition mode
- Randomizer Module — shuffle mode selecting a surprise game from the hub
- Time & Tries Limit — pressure mode with countdown timer and attempt constraints
- Account Management — edit username, reset scores, or delete account with confirmation step
- Data Persistence — custom file I/O system saving and loading all user data between runs

TECH STACK: C++ · Object-Oriented Programming (OOP) · Custom Data Structures & Algorithms · File I/O · Terminal UI Design

💡 **WHAT I LEARNED:** Architecture matters more than features. A well-structured codebase lets you add the sixth game in an afternoon. A poorly structured one means rewriting everything. This was the project that made that lesson permanent.

SPECIAL EPISODE — SUMMER 2024 | WORK EXPERIENCE

Software Development Intern @ Sense.ai

97% Match

First industry deployment. First real client. First production website. No professor to check the work.

- **Period:** Jun 2024 – Aug 2024 (2 months)
- **Company:** Sense.ai — AI-driven productivity and personal growth platform
- **Type:** Internship · On-site / Remote · Lahore, Pakistan
- **Genre:** Front-End Development · Professional Experience · Production Deployment
- **Status:** ✅ Completed

THE PLOT

The bridge between academia and industry. At Sense.ai, the assignment was to design and build their official company website from scratch — a site representing a real AI startup's brand to real customers. No sample solution, no marking rubric, no retry button. The full design-to-deployment cycle was owned end-to-end: brand identity workshops, UI decisions, front-end implementation, cross-functional content integration, and performance optimization before launch.

KEY CONTRIBUTIONS

- Website Design & Development — full website built from zero: wireframes → design → development → live deployment
- Front-End Engineering — responsive, accessible, performant UIs in HTML5, CSS3, and JavaScript
- UX/UI Design — user experience decisions rooted in Sense.ai's brand vision and target user personas
- Cross-Functional Collaboration — coordinated with content, product, and leadership teams to integrate messaging and case studies
- Performance Optimization — asset compression, code refactoring, and lazy loading; 15% reduction in page load time
- Engagement Impact — UX improvements drove a measurable 20% increase in user engagement post-launch

TECH STACK: HTML5 · CSS3 · Bootstrap · JavaScript (Vanilla) · Responsive Web Design · Git · GitHub

💡 **WHAT I LEARNED:** Industry moves faster than university. A decision that takes a week in a group project gets made in an afternoon with a real client. Learning to make confident, well-reasoned decisions quickly is the real internship outcome.

📺 SEASON 3 — THE DATABASE ARC

Fall 2024

Year Two begins. OOP, databases, and a real client. The semester that introduced persistence, normalization, and the quiet satisfaction of a perfectly structured schema. Also: first teaching role.

EPISODE 01 — Banking Management System

96% Match

Secure. Scalable. Object-Oriented. What banking looks like when C++ meets MySQL.

- **Period:** Nov 2024 – Dec 2024
- **Course:** Object-Oriented Programming (OOP) — Semester 3
- **Genre:** FinTech · Backend Architecture · C++ · Database Integration · OOP
- **Status:** ✅ Completed

THE PLOT

A comprehensive financial simulation engine bridging volatile runtime code with permanent relational storage. The system simulates end-to-end banking operations — from secure account creation and authentication through live transaction tracking to persistent record storage — demonstrating full command of OOP principles including Polymorphism, Inheritance, and Encapsulation. The architectural challenge was connecting a C++ application layer to a live MySQL database in real time. Every transaction had to be written immediately and retrieved correctly on next login. No data lost between sessions. No account accessible without proper authentication.

KEY FEATURES

- Secure Authentication — encapsulated login system with password management and account lockout after failed attempts
- Account Management — create, view, update, and close accounts with role verification
- The Ledger — real-time deposit and withdrawal processing with instant balance update
- Transaction History — complete timestamped log of all operations, retrievable per account
- C++ to MySQL Bridge — live bidirectional data flow between application and relational database

- OOP Architecture — class hierarchy: Account (base) → SavingsAccount, CurrentAccount (derived)
- Polymorphism — overridden interest calculation and overdraft rules per account type
- Error Handling — robust input validation and exception management throughout
- Admin Mode — privileged access tier for system-wide account oversight and reports

TECH STACK: C++ · MySQL · OOP (Polymorphism · Inheritance · Encapsulation) · MySQL Connector/ODBC · File I/O

💡 **WHAT I LEARNED:** Connecting code to a database changes everything. Data stops being a variable and starts being permanent. That shift — from ephemeral to persistent — is fundamental to building real software.

EPISODE 02 — Skinsations Clinic Database

97% Match

Real-world data infrastructure for a real client. Not a simulation — a production-grade design.

- **Period:** Oct 2024 – Jan 2025
- **Course:** Database Systems — Semester 3
- **Genre:** Healthcare Tech · Real-World Client · Relational Database Design · ERD
- **Status:** ✅ Completed · Client-Delivered

THE PLOT

Most database projects are textbook exercises. This one was different. Skinsations is a real aesthetic clinic in Lahore — and this project involved working with actual stakeholders to understand real operational workflows, then translating those into a fully normalized, production-grade relational database. Stakeholder interviews mapped the clinic's actual data flows: patient registration, appointment booking, treatment logging, billing, inventory. Every design decision emerged from a real requirement. The result was a schema that didn't just work in theory — it worked for the clinic's actual operations.

KEY FEATURES

- Patient Profile Management — complete records with medical history, contact details, and treatment preferences
- Appointment Scheduling — booking, rescheduling, cancellation with conflict detection and calendar view
- Treatment Logs — per-visit records including treatments administered, products used, and clinician notes
- Staff & Clinician Records — staff profiles with role-based data access control

- Billing & Payment Tracking — invoice generation, payment status, and outstanding balance management
- Inventory Management — product and supply tracking linked to treatment records
- Normalized Schema (3NF) — full third normal form to eliminate redundancy and ensure data integrity
- Entity-Relationship Diagram — complete ERD with all entities, attributes, and cardinalities documented
- Optimized SQL Queries — complex scripts for fast retrieval, reporting, and data insertion
- Client Consultation Process — requirements gathered through direct stakeholder interviews with clinic staff

TECH STACK: MySQL · Relational Database Design · Entity-Relationship Diagrams (ERD) · Schema Normalization (3NF) · SQL Scripting

💡 **WHAT I LEARNED:** Client work is a completely different discipline. A theoretically perfect schema the client can't understand is a failed project. Translation — from human workflow to relational structure — is half the job.

BONUS EPISODE — FALL 2024 | LEADERSHIP

Teaching Assistant — Applied Physics Lab

Note: I have also been TA for AI Lab, ICT Lab & Programming Fundamentals So find a way to group them up together

When you know something well enough to teach it, you actually know it.

- **Period:** Oct 2024 – Jan 2025
- **Organization:** Beaconhouse National University — Applied Physics Lab
- **Genre:** Leadership · Education · Technical Communication
- **Status:** ✅ Completed

THE PLOT

Selected by faculty to support the Applied Physics Lab as a Teaching Assistant — one of the earliest leadership roles in the degree. Responsibilities included facilitating hands-on lab sessions, guiding experimental procedures, troubleshooting equipment issues in real time, and delivering structured written feedback on 100+ submitted lab reports across the semester.

Beyond the technical side, this role developed two skills no coding project can teach: explaining complex ideas simply, and the patience to meet learners exactly where they are.

KEY RESPONSIBILITIES

- Lab Facilitation — led weekly hands-on lab sessions guiding students through experimental procedures
- Real-Time Troubleshooting — diagnosed and resolved equipment, measurement, and calculation issues during live sessions
- Report Grading — assessed 100+ lab reports per semester with detailed, constructive written feedback
- Academic Support — held supplementary help sessions for students needing additional guidance
- Curriculum Contribution — collaborated with faculty on session planning and material improvement

💡 **WHAT I LEARNED:** Teaching is the best test of your own understanding. If you can explain a concept clearly to someone seeing it for the first time, you actually know it. If you can't — you have more to learn, and that's valuable information.

📺 SEASON 4 — THE AI AWAKENING

Spring 2025

The breakout season. LLMs, RAG, web scraping, a cross-department collaboration, an IEEE-published research paper, and two industry awards. The semester that changed everything.

EPISODE 01 — Finnovation — Intelligent Finance Chatbot

96% Match

Your money, but make it AI. Context-aware, retrieval-augmented, actually useful.

- **Period:** Jan 2025
- **Organization:** 4X Pillars
- **Genre:** FinTech · AI · NLP · Conversational AI · RAG · LLM Integration
- **Status:** ✅ Completed

THE PLOT

Finnovation is an AI-powered finance assistant that answers real financial questions with context, accuracy, and up-to-date information. The core challenge with financial chatbots is staleness: a model trained on static data becomes unreliable the moment market conditions change. Finnovation solves this with Retrieval-Augmented Generation (RAG) — a pipeline that dynamically retrieves relevant documents and live data at query time, then passes them as context to an LLM to generate grounded, accurate responses. Every component was engineered for financial domain accuracy, where a hallucinated figure can have real consequences.

KEY FEATURES

- RAG Pipeline — complete retrieval-augmented generation architecture: ingest → chunk → embed → retrieve → generate
- LLM Integration — large language model backbone with conversation memory for coherent multi-turn dialogue
- NLP Query Understanding — intent recognition and named entity extraction for financial terminology
- Context-Aware Responses — multi-turn conversation history preserved for coherent follow-up handling
- Real-Time Data Integration — external API connections to live financial data sources
- Grounded Generation — retrieved documents passed as context to prevent hallucination on factual data
- Domain Prompt Engineering — system prompts optimized for financial assistant behavior and safety

TECH STACK: Python · NLP · Retrieval-Augmented Generation (RAG) · LLMs · HuggingFace Transformers · Vector Embeddings · External APIs

💡 **WHAT I LEARNED:** RAG is the gap between an AI that sounds smart and an AI that actually is smart. Building the pipeline yourself makes it obvious why grounding matters — especially in high-stakes domains where a wrong figure isn't just an error, it's potentially harmful.

EPISODE 02 — Grantify — Automated Research Grant Monitor

94% Match 🏆

Eliminating the manual grind. Automated grant discovery for a real university department.

- **Period:** Mar 2025 – Apr 2025

- **Organization:** BNU Office of Research & Partnerships (ORIP)
- **Genre:** Automation · Web Scraping · Research Infrastructure · Real Client · AI
- **Status:** ✅ Completed · Best Project Award — AI Category, BNU Project Display 2025
- 🏆 Best Project Award — AI Category, BNU Project Display 2025 (Usman Asif, CEO — Devsinc)

THE PLOT

BNU's Office of Research & Partnerships was manually monitoring dozens of grant databases, funding bodies, and government portals — hours of weekly effort that still missed deadlines. Grantify automates the entire pipeline: scrapes target sources on a schedule, parses new listings for title, deadline, eligibility, and funding amount, filters by relevance to BNU's research areas, and fires structured notifications to the relevant staff. A multi-hour weekly task becomes a zero-effort automated workflow that never misses a deadline. The project won Best Project in the AI Category at BNU's 2025 Project Display, presented by Usman Asif, CEO of Devsinc.

KEY FEATURES

- Automated Web Scraping — scheduled crawling of multiple grant databases and government funding portals
- Smart Parsing Engine — structured extraction of grant title, issuer, deadline, eligibility, funding amount, and application URL
- Relevance Filtering — keyword and domain matching to surface only grants relevant to BNU research areas
- Notification Pipeline — automated email alert system delivering structured grant summaries to ORIP staff
- Deduplication System — prevents repeat notifications for already-seen grants across scraping cycles
- Configurable Scheduling — daily / weekly run frequency with comprehensive audit logging
- Error Handling — graceful recovery when target sites change structure or go temporarily offline

TECH STACK: Python · Beautiful Soup 4 · Requests · Schedule / Cron · Email Automation (smtplib) · Data Parsing · Web Scraping

👥 **COLLABORATORS:** Hania Atta, Abdul Hadi Jalal

🎓 **MENTOR:** Dr. Usman Nazir

💡 **WHAT I LEARNED:** Automation is not about replacing humans — it's about freeing them for the work that actually requires a human. Grantify didn't eliminate the ORIP team's jobs. It gave them their Mondays back.

EPISODE 03 — AfterLight — Narrative Browser-Based Game

93% Match 🏆

Arts meets engineering. A dystopian world, child-bots, and a cross-department collaboration that won the room.

- **Period:** Mar 2025 – Apr 2025
- **Organization:** BNU Connect — SCIT × SVAD Inter-Department Collaboration
- **Genre:** Game Development · Narrative Design · Cross-Disciplinary · Browser-Based · UI/UX
- **Status:** ✅ Completed · Award of Excellence — BNU Connect 2025
- 🏆 Award of Excellence — BNU Connect 2025

THE PLOT

AfterLight is a narrative browser-based game set in a dystopian world where humanity no longer exists. AI systems are training child-bots by revisiting the downfall of human civilization — a story about the consequences of human choices. Built as a graduating SVAD (School of Visual Arts & Design) student's thesis project, it required genuine collaboration between computer science and visual arts to bring the vision to life. The challenge was translational: encoding a rich artistic and narrative vision in code without losing its meaning. Every UI decision, every interaction mechanic, every transition had to serve the story. The Award of Excellence at BNU Connect 2025 recognized what happens when two disciplines genuinely collaborate rather than just coexist.

KEY CONTRIBUTIONS

- C++ Game Logic — core mechanics engine: player state management, narrative branching, win/loss condition handling
- UI/UX Design — interface matching the dystopian aesthetic: dark palette, atmospheric typography, minimal HUD
- Browser Integration — C++ game logic bridged to web-facing front-end for accessible browser delivery
- Narrative Mechanics — interactive decision trees advancing the story based on player choices
- Dynamic Interactions — animated transitions, event-driven interactions, responsive layout
- Cross-Department Coordination — sustained collaboration between SCIT and SVAD for unified vision

TECH STACK: C++ · JavaScript · HTML5 · CSS3 · UI/UX Design · Browser-Based Game Architecture

👥 **COLLABORATORS:** SVAD Thesis Student (Collaborating Artist & Narrative Designer)

💡 **WHAT I LEARNED:** The best technical work disappears into the experience. In AfterLight, the goal was for the player to feel the story — not notice the code. Learning to subordinate technical choices to creative purpose makes everything better.

SPECIAL EPISODE — SPRING 2025 | RESEARCH PUBLICATION

AzureIQ-RAG — IEEE ICIT 2025 Published Paper

100% Match 📄

From coursework to conference proceedings. A published IEEE paper as a third-year undergraduate.

- **Period:** Spring 2025
- **Venue:** IEEE ICIT 2025 — International Conference on Industrial Technology
- **Title:** AzureIQ-RAG: Azure Powered Intelligent Query and Retrieval-Augmented Generation
- **Genre:** AI Research · RAG · Azure Cloud · Academic Publication · Peer-Reviewed
- **Status:** ✅ Published — IEEE ICIT 2025

THE PLOT

A research paper on AzureIQ-RAG — an Azure-powered intelligent query system built on Retrieval-Augmented Generation — was accepted and published at IEEE ICIT 2025. Getting research accepted at an IEEE conference as a third-year undergraduate is not a common achievement. The paper explores how Azure's cloud infrastructure can be leveraged to build production-grade RAG pipelines — combining Azure Cognitive Search, Azure OpenAI, and storage services into an intelligent, scalable query-answering system, addressing real limitations of static LLMs in enterprise knowledge management.

KEY CONTRIBUTIONS

- Novel RAG Architecture — AzureIQ-RAG pipeline integrating Azure Cognitive Search + Azure OpenAI for grounded enterprise query resolution
- Experimental Evaluation — benchmarking of retrieval accuracy, response quality, and latency across document corpus sizes
- Azure Cloud Integration — full implementation on Azure infrastructure for enterprise scalability and deployment readiness

- Academic Writing — complete research paper written to IEEE conference publication standards
- Peer Review Process — paper reviewed and accepted through formal IEEE peer review

TECH STACK: Python · Azure Cognitive Search · Azure OpenAI · Retrieval-Augmented Generation · Cloud Architecture · NLP

💡 **WHAT I LEARNED:** Writing a research paper forces a precision that code alone doesn't require. Every claim needs evidence. Every design decision needs justification. The discipline of academic writing makes the engineering thinking sharper.

SPECIAL EPISODE — SPRING 2025 | WORK EXPERIENCE

Research & Outreach Analyst @ Digital Ocean WET Program

91% Match

Data-driven impact for Pakistan's digital transformation. Early morning buses and cups of chai included.

- **Period:** Apr 2025 – Present
- **Organization:** Digital Ocean Women Entrepreneurs Training (WET) Program — BNU × DigitalOcean
- **Type:** Part-Time Analyst · Field + Office · Lahore and Regional Sites, Pakistan
- **Genre:** Research · Data Analysis · Social Impact · Women's Empowerment · Field Outreach
- **Status:** 🟢 Ongoing

THE PLOT

The WET Program trains women entrepreneurs across Pakistan in digital transformation — helping them adopt technology to grow their businesses. As Research & Outreach Analyst, the role is to measure whether the training actually works. This means field visits to regional training sites, stakeholder interviews with participating entrepreneurs, data collection on pre/post-training business outcomes, and reporting findings back to program leadership. It is equal parts researcher, analyst, and community liaison — and it involves genuine conversations with women whose resilience is remarkable.

KEY RESPONSIBILITIES

- Data Collection — qualitative interviews and quantitative surveys at regional sites including University of Okara and FCCU Lahore
- Impact Analysis — measuring pre/post-training outcomes: digital adoption rates, business revenue changes, tool usage
- Stakeholder Engagement — ongoing communication with 50+ participating entrepreneurs to assess challenges and progress
- Field Visits — on-site presence at training events across Punjab province
- Report Writing — structured impact assessment reports delivered to program leadership and BNU faculty
- Strategic Recommendations — data-backed proposals for curriculum adjustments and targeted participant support
- Photography & Documentation — visual documentation of program events for institutional records

👥 **COLLABORATORS:** Team Lead: Hafiz Muhammad Abubakar · Fellow Analyst: Tayyab Mahmood

🎓 **MENTOR:** Dr. Usman Nazir, Mr. Nouman Ali

💡 **WHAT I LEARNED:** Technology solves the wrong problem if you don't understand the people it's supposed to serve. Sitting with a woman entrepreneur in Okara and listening to her actual barriers — not assumed ones — changes how you think about what to build and why.

📺 SEASON 5 — THE FLAGSHIP SEASON ⭐

Fall 2025

The season everything leveled up. Deepfake detection. Pakistan's first women's safety super-app. Pakistan's first APAC Cyber Clinic. A hackathon to lead. This is the one.

EPISODE 01 — Digital Forensics: Deepfake Detection via CNNs

99% Match 🔥

Because not everything you see is real — and someone has to build the system that catches the fakes.

- **Period:** Dec 2025 – Jan 2026
- **Organization:** Beaconhouse National University — Deep Learning Research Project
- **Genre:** AI · Digital Forensics · Computer Vision · Deep Learning · Explainable AI · Cybersecurity
- **Status:** ✅ Completed

THE PLOT

Deepfakes are one of the defining threats of the AI era. As generative models — GANs and Diffusion Models — become accessible to anyone with a GPU and an internet connection, the ability to detect AI-generated imagery becomes critical infrastructure for journalism, law enforcement, and social media platforms. This project builds that detection system.

An end-to-end deep learning pipeline was designed and implemented to classify real photographs versus AI-generated images with the precision required for forensic-grade verification. Three architectures were built, trained, and benchmarked — not to find a single winner, but to understand the full trade-off landscape: accuracy vs. efficiency, server-side vs. edge deployment, semantic robustness vs. lightweight portability.

Grad-CAM was integrated to confirm the model detects actual generative artifacts — the subtle texture anomalies GANs and diffusion models leave behind — not facial identity patterns. This is the difference between a model that works and a model that can be trusted in court.

ARCHITECTURE COMPARISON

- **Baseline CNN** — custom architecture identifying high-frequency GAN pixel artifacts: spatial noise patterns, checkerboard artifacts, and texture inconsistencies that distinguish synthetic from real at the signal level
- **Xception (Fine-Tuned Transfer Learning)** — depthwise separable convolutions capture rich semantic features. Fine-tuned on the forensic dataset for server-side deployment; optimized for precision to minimize false positives in forensic contexts
- **MobileNetV2 (Edge-Optimized)** — lightweight architecture proving real-time deepfake detection on mobile and edge hardware. Forensic tools don't require a data centre.

ENGINEERING HIGHLIGHTS

- Memory-Efficient Data Pipeline — custom lazy-loading with manual path shuffling to prevent data leakage on hardware-constrained environments

- Grad-CAM (Explainable AI) — gradient-weighted class activation maps confirming detection of hair-strand irregularities and background texture artifacts — NOT facial identity
- Training Optimization — Adam optimizer, Binary Crossentropy loss, ReduceLROnPlateau scheduling, early stopping to prevent overfitting
- Transfer Learning Pipeline — ImageNet weights fine-tuned on forensic dataset; upper layers unfrozen in stages to prevent catastrophic forgetting
- Comparative Benchmarking — precision, recall, F1-score, AUC-ROC evaluated across all three architectures

TECH STACK: Python · TensorFlow · Keras · CNN Design · Transfer Learning (Xception · MobileNetV2) · Grad-CAM (XAI) · NumPy · Pandas · Matplotlib · Scikit-learn

💡 **WHAT I LEARNED:** The gap between 'it works' and 'it can be trusted' is enormous in forensic AI. A model with 97% accuracy you can't explain is less useful in a legal context than an 89% model with interpretable, auditable decision logic. Explainability is not a nice-to-have — it is the product.

EPISODE 02 — Umeed-e-Zann — Women's Safety & Empowerment App

98% Match 💜

Pakistan's first digital ecosystem built by women, for women. Hope, reimagined through technology.

- **Period:** Fall 2025
- **Event:** BNU Fall 2025 Project Display — Received overwhelming response
- **Meaning:** Umeed = Hope · Zann = Woman · وجود زن سے ہے تصویر کائنات میں رنگ
- **Genre:** Mobile App · Flutter · Social Impact · AI Integration · Women's Tech · Safety
- **Status:** ✅ Presented at BNU Fall 2025 Project Display

THE PLOT

Umeed-e-Zann answers a question that shouldn't still need answering in 2025: where does a woman in Pakistan go when she needs safety, legal guidance, health information, mental health support, or simply a community that understands? Too often the answer is: nowhere good enough. This platform was built to change that.

Named for Umeed (hope) and Zann (woman), and rooted in the verse وجود زن سے ہے تصویر کائنات میں رنگ — the world takes its colour from the existence of women — Umeed-e-Zann was designed from the first screen to feel like a safe space: calming, reassuring, judgment-free.

Every feature exists because a real woman in Pakistan has a real unmet need for it. The response at BNU's Fall 2025 Project Display was overwhelming — judges, faculty, and students who explored the app understood the vision immediately.

PLATFORM PILLARS & FEATURES

- Safety & Emergency Support — one-tap SOS with live location sharing to trusted contacts, emergency service directory, and safe route guidance
- Women's Health & Wellness — expert-verified health information covering reproductive health, mental wellness, and preventive care in Urdu and English
- Legal Aid with AI Assistance — AI-powered legal guidance translating Pakistani law into plain accessible language; case type identification and referral to verified legal aid organisations
- Mental Health Support — self-assessment tools, curated coping resources, and directory of licensed mental health professionals
- Community Spaces — moderated peer-to-peer support forums; topic-based communities for shared experiences
- Women-Led Marketplace — platform for women entrepreneurs to list products and services
- Learning, Mentorship & Growth — curated skill-building courses, mentorship matching, and career development resources
- Safe Design Language — UI/UX crafted for psychological safety: soft palette, clear typography, non-intimidating navigation
- Bilingual Interface — full Urdu and English throughout the application

TECH STACK: Flutter · Dart · AI Integration (Legal Aid Module) · REST APIs · Mobile UI/UX Design · State Management

👥 **COLLABORATORS:** Hania Atta, Abdul Hadi Jalal, Zoya Khan

🎓 **MENTOR:** Mr. Asim Irshad

💡 **WHAT I LEARNED:** The most important thing you can do as a developer is care about the person on the other side of the screen. Every feature, every design decision, every word in the UI exists to make someone feel less alone. That is what this degree is for.

SPECIAL EPISODE — FALL 2025 | WORK EXPERIENCE

Cybersecurity Trainee @ InnovAtrium — APAC Cyber Clinic

99% Match 🔒

Pakistan's first. Google.org × The Asia Foundation × Global Cyber Alliance. History being made.

- **Period:** Nov 2025 – Present
- **Organization:** InnovAtrium — Google.org · The Asia Foundation · Global Cyber Alliance
- **Type:** Contract · On-site · Lahore, Pakistan
- **Scale:** 100 students trained over 2 years · 300 SMEs supported with cybersecurity protection
- **Genre:** Cybersecurity · Industry Training · SME Protection · National Initiative
- **Status:** 🟢 Ongoing

THE PLOT

Not every internship is a historic first. This one is. Selected for Pakistan's inaugural APAC Cyber Clinic — a competitive, nationally significant programme co-designed by Google.org, The Asia Foundation, and the Global Cyber Alliance to build Pakistan's cybersecurity capacity at the SME level. Training is delivered by industry practitioners — including Mr. Uzair Bashir and Nabeel Ahsan — with genuine field experience. The curriculum covers network security, phishing detection, data protection, digital hygiene for organizations, and practical threat response. Beyond individual skill-building, fellows collaborate to design and deliver cybersecurity training modules for partner institutions. This isn't a certificate course — it's a deployment.

TRAINING SCOPE

- Network Security — perimeter defence, traffic analysis, firewall configuration, and infrastructure hardening
- Threat Intelligence — identifying, categorizing, and tracking real-world cyber threats and threat actor profiles
- Phishing Attack Detection — recognizing social engineering vectors; analyzing phishing emails, domains, and credential harvesting
- Data Protection Practices — encryption standards, secure data handling, access control, and privacy compliance
- Digital Hygiene for SMEs — practical security posture improvement for businesses with limited IT resources
- Cybersecurity Training Design — developing modules for students at SMC and partner institutions
- Scenario-Based Exercises — hands-on workshops simulating incident response and threat mitigation
- SME Vulnerability Exposure — real-world cybersecurity risk landscapes of Pakistani small-medium enterprises

👥 **COLLABORATORS:** Cohort of 100 students — Pakistan's first APAC Cyber Clinic

🎓 **MENTORS:** Mr. Uzair Bashir, Nabeel Ahsan (InnovAtrium Cybersecurity Practitioners)

💡 **WHAT I LEARNED:** Cybersecurity is not a product you install. It is a culture you build. Watching SME owners understand why their password habits matter — and seeing that shift their behaviour — is more impactful than any technical configuration you could deploy for them.

BONUS EPISODE — FALL 2025 | LEADERSHIP

Head of Neural Nova — BTech Fest, BNU

Run the room. Set the theme. Build a better tomorrow. This is what AI leadership looks like.

- **Period:** 2025
- **Event:** BTech Fest — BNU's flagship annual technology hackathon
- **Track:** Neural Nova — Premier AI/ML Competition Category
- **Theme:** Build A Better Tomorrow
- **Genre:** Leadership · AI/ML · Hackathon Direction · Event Management
- **Status:** ✅ Completed

THE PLOT

Appointed Head of Neural Nova — the AI/ML flagship track at BNU's premier hackathon, BTech Fest. The mandate went far beyond logistics: define what AI innovation at BNU looks like in 2025. The theme 'Build A Better Tomorrow' was chosen as a deliberate pushback against AI competitions that reward technical sophistication without purpose. The brief was explicit: don't build something impressive — build something that matters. Whether solutions addressed climate change, healthcare access, digital privacy, or everyday community challenges — the standard was impact, not spectacle.

KEY RESPONSIBILITIES

- Track Direction — defined Neural Nova's scope, rules, judging criteria, and thematic vision for 2025
- Theme Curation — selected and justified 'Build A Better Tomorrow' to elevate social impact AI submissions
- Participant Management — onboarding, briefings, and Q&A for all AI/ML category participants
- Judging Framework — evaluation rubric balancing technical execution, innovation, and real-world applicability
- Mentor Coordination — coordinated with industry mentors and faculty judges
- Problem Statement Design — crafted challenge prompts across climate, healthcare, privacy, and community domains

💡 **WHAT I LEARNED:** Leadership in a technical context is not about knowing the most. It is about creating the conditions where other people can do their best work — and then holding them to a standard worth meeting.

📺 SEASON 6 — NOW STREAMING ▶

Spring 2026

The season currently being written. Production MERN stack at a real AI startup. More episodes dropping. Check back soon.

SPECIAL EPISODE — SPRING 2026 | WORK EXPERIENCE

Full-Stack Development Intern @ Osso.ai

98% Match

Production MERN stack. Real APIs. Real users. No safety net. This is what shipping feels like.

- **Period:** Sep 2025 – Nov 2025 (3 months)
- **Organization:** Osso.ai — AI-driven solutions platform, Islamabad, Pakistan
- **Type:** Internship · Remote
- **Genre:** Full-Stack Development · MERN Stack · API Engineering · AI Integration · Production
- **Status:** ✅ Completed

THE PLOT

Three months building real things for real users at a live AI startup. At Osso.ai, the full MERN stack was in play from day one: MongoDB databases, Express.js and Node.js backends, and React frontends, all connected through secure RESTful APIs. Every line of code was production-bound. The scope went beyond front-end: designing database schemas for scalability, engineering authenticated API endpoints, integrating AI components into the application layer, and optimizing backend performance under real load.

KEY CONTRIBUTIONS

- Full-Stack Application Development — designed and deployed scalable MERN stack applications end-to-end
- RESTful API Engineering — built and maintained secure, JWT-authenticated APIs with proper error handling and rate limiting
- React Front-End Development — component-based UI architecture with hooks, state management, and responsive design
- Node.js / Express.js Backend — server architecture with middleware, routing, and request lifecycle management
- MongoDB Schema Design — document modelling optimized for query performance and horizontal scalability
- AI/ML Integration — embedded intelligent, data-driven features via AI model API connections
- Backend Optimization — database indexing, query optimization, and caching strategies for improved throughput
- Code Quality & Version Control — code reviews, documentation, production-grade Git workflows

TECH STACK: MongoDB · Express.js · React · Node.js (MERN Stack) · RESTful API Design · JWT Authentication · Git · GitHub · AI/ML API Integration

💡 **WHAT I LEARNED:** Production is humbling. A bug in a university project gets a comment from a professor. A bug in production gets a message from a user. The feedback loop is faster, more honest, and infinitely more motivating. Ship carefully. Ship often. Fix fast.

⏳ *Season 6 episodes currently in development. More projects dropping Spring 2026. Watch this space.*

🏆 AWARDS & RECOGNITION

The Trophies Wall — Every badge, every honour, with full context.

🏆 AI INNOVATION DISTINCTION AWARD

Recognized by an industry leader for outstanding contribution to Artificial Intelligence.

- **Event:** Annual Project Showcase 2025 — Beaconhouse National University
- **Presented By:** Usman Asif, CEO — Devsinc

- **Date:** May 2025

Received for the Grantify project — an automated grant monitoring system that leveraged web scraping and AI automation to solve a real problem for BNU's research office. Presented by the CEO of one of Pakistan's leading software companies. Recognition from an industry CEO at a university showcase signals that the work had value beyond the academic context.

🏆 BEST PROJECT AWARD — AI CATEGORY

Grantify wins the AI category at BNU's Project Display 2025.

- **Event:** BNU Project Display 2025
- **Project:** Grantify — Automated Research Grant Monitoring System
- **Date:** Spring 2025

Awarded Best Project in the AI Category at BNU's 2025 Project Display, competing against the full cohort of AI/ML submissions. Recognized for technical execution, genuine utility for a real client, and measurable impact on administrative workflows.

🏆 AWARD OF EXCELLENCE — BNU CONNECT 2025

Excellence in interdisciplinary collaboration. AfterLight brings CS and Visual Arts together.

- **Event:** BNU Connect 2025
- **Project:** AfterLight — Narrative Browser-Based Game
- **Date:** Summer 2025

Presented for AfterLight — developed in collaboration between BNU's SCIT and SVAD schools. Recognized for outstanding integration of computer science and visual arts disciplines, producing an experience neither could have created independently.

🏆 DANDELIONS — THE DOUBLE WINNER

Two awards. One semester. The only project in cohort history to win both simultaneously.

- **Awards:** Best Team Project (Web Development) · Best Team Project (Programming)
- **Presented By:** Ayub Ghauri — NETSOL Technologies
- **Date:** June 2024

Swept two categories at the 2024 BNU Project Showcase. Winning both is notable because they evaluate different dimensions: front-end quality and UX vs. code logic and structure. Dandelions excelled on both simultaneously.

📚 DEAN'S HONOR LIST — ONGOING SERIES

Consistent top-tier academic performance. Every semester without exception.

- **Period:** Feb 2024 – Present (every semester)
- **Issuer:** Dr. Shafaat Bazaaz — Beaconhouse National University
- **Also:** Merit Scholarship — maintained throughout the full undergraduate degree

Earned every semester since starting the CS programme, alongside the Merit Scholarship. Consistent performance across every course while simultaneously running internships, projects, a published IEEE paper, and leadership roles. Current CGPA: 3.79.

🎖️ AI RESEARCHER — CENTRE OF AI RESEARCH (CAIR), BNU

- **Organization:** Centre of AI Research (CAIR), Beaconhouse National University
- **Date:** 2025

Formally designated as an AI Researcher within BNU's Centre of AI Research — institutional recognition of applied research contributions during the third year. Reflects the IEEE publication and applied research in the CNN deepfake detection pipeline and AzureIQ-RAG.

🎓 CAMPUS AMBASSADOR — VICE CHANCELLOR'S OFFICE

- **Appointed By:** Vice Chancellor Moeed Yusuf — Beaconhouse National University
- **Date:** Fall 2024

Selected as an official BNU Campus Ambassador — formal liaison between student body and university administration, appointed by the Vice Chancellor. One of a small cohort selected university-wide.

🏅 ACADEMIC HIGH ACHIEVER — BAHRIA TOWN COLLEGE

- **Issuer:** Bahria Town College, Lahore
- **Date:** 2023

Graduated with Academic High Achiever distinction — exceptional pre-university performance that preceded entry into BNU's Computer Science programme. The foundation for everything that followed.

📜 CERTIFICATIONS

Docuseries & Specializations — The continuing education catalogue.

▸ GOOGLE MACHINE LEARNING CRASH COURSE

A deep dive into the architecture of modern AI. Eleven badges. One obsession.

- **Issuer:** Google Developers Group
- **Completed:** July 2025
- **Format:** Standalone Intensive — 11 Distinct Learning Badges

BADGES EARNED:

1. Introduction to Machine Learning — supervised learning fundamentals, model evaluation, bias-variance tradeoff
2. Problem Framing — translating real-world problems into well-defined ML problem statements
3. Data Preparation — feature engineering, data cleaning, handling class imbalance and missing data
4. Logistic Regression — binary and multiclass classification, probability calibration, ROC curves
5. Neural Networks — architecture design, activation functions, layers, forward propagation
6. Training Neural Networks — backpropagation, gradient descent variants, learning rate schedules, regularization
7. ML Fairness — bias identification, fairness metrics, responsible AI principles and evaluation
8. Decision Forests — random forests, gradient boosted trees, ensemble methods, feature importance
9. Large Language Models — transformer architecture, tokenization, prompt design, capabilities and limitations
10. Introduction to Generative AI — generative model taxonomy: GANs, diffusion models, VAEs
11. Responsible AI Practices — deployment ethics, model monitoring, harm prevention, AI accountability

#MachineLearning #NeuralNetworks #LLMs #GenerativeAI #ResponsibleAI #DeepLearning

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▸ GOOGLE IT AUTOMATION WITH PYTHON

The complete 6-episode professional specialization on Python automation.

- **Issuer:** Google via Coursera
- **Completed:** July 2024
- **Format:** Professional Certificate Specialization — 6 Courses

EPISODE BREAKDOWN:

- Ep 01: Crash Course on Python (May 2024) — syntax, data types, functions, OOP fundamentals, scripting foundations
- Ep 02: Using Python to Interact with the OS (Jul 2024) — file system operations, process management, text processing, bash scripting
- Ep 03: Git and GitHub (Jun 2024) — version control, branching, merging, pull requests, collaborative workflows
- Ep 04: Troubleshooting and Debugging (Jul 2024) — root cause analysis, systematic debugging, performance profiling
- Ep 05: Configuration Management and the Cloud (Jul 2024) — infrastructure as code, Puppet, cloud resource management, scaling
- Ep 06: Automating Real-World Tasks with Python (Jul 2024) — API integration, data pipelines, final capstone project

#Python #DevOps #Automation #Git #CloudInfrastructure #Scripting

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▸ CS50: INTRODUCTION TO COMPUTER SCIENCE

The one that started it all. Harvard's legendary gateway to computer science.

- **Issuer:** Harvard University via edX
- **Completed:** 2023
- **Format:** Standalone Course — Harvard CS50

Completed Harvard's globally acclaimed CS introduction — covering algorithms, data structures, memory management, and problem-solving across C, Python, SQL, and web development. The first encounter with pointers, hash tables, and computational complexity. The course that confirmed computer science was the right path.

#CS50 #Harvard #Algorithms #C #Python #SQL #DataStructures

WORK EXPERIENCES

Cybersecurity Trainee InnovAtrium · Contract · On-site Nov 2025 – Present · 6 months Lahore, Punjab, Pakistan

Selected as part of Pakistan's first APAC Cyber Clinic cohort — a programme co-designed with Google.org, The Asia Foundation, and the Global Cyber Alliance to train 100 students and support 300 SMEs with cybersecurity protection. Receiving hands-on training in network security, threat intelligence, phishing detection, and security operations under expert industry mentorship. Collaborating with cohort fellows to develop and deliver cybersecurity awareness modules for partner institutions across Pakistan.

Full-Stack Development Intern Osso.ai · Internship · Remote Sep 2025 – Nov 2025 · 3 months Islamabad, Pakistan

Built and deployed scalable full-stack web applications using the MERN stack (MongoDB, Express.js, React, Node.js) in a production environment. Engineered and maintained secure RESTful APIs with JWT authentication. Integrated AI/ML components into web architecture and optimized database schemas and backend performance for scalability and throughput.

Research & Outreach Analyst Digital Ocean WET Program · Part-Time · Field + Office Apr 2025 – Present Lahore, Punjab, Pakistan

Collected and analyzed qualitative and quantitative data at regional training sites across Punjab — including University of Okara and FCCU Lahore — to evaluate digital transformation programme effectiveness for women entrepreneurs. Conducted 50+ stakeholder interviews, delivered impact assessment reports to program leadership, and proposed data-backed curriculum improvements.

Teaching Assistant — AI Lab, ICT Lab, Programming Fundamentals Lab & Applied Physics Lab Beaconhouse National University · Part-Time · On-site Oct 2024 – Present Lahore, Punjab, Pakistan

Served as Teaching Assistant across four labs and courses over multiple semesters — Applied Physics Lab (Oct 2024 – Jan 2025), ICT Lab, Programming Fundamentals Lab, and AI Lab (ongoing). Facilitated weekly lab sessions, guided students through practicals and coding exercises, graded 100+ reports and assignments per semester with written feedback, and held supplementary support sessions for students needing additional help.

Software Development Intern Sense.ai · Internship · On-site Jun 2024 – Aug 2024 · 2 months Lahore, Punjab, Pakistan

Designed and built the company's official website from scratch — wireframes through to live deployment. Developed responsive front-end UIs in HTML5, CSS3, and JavaScript. Achieved a 15% reduction in page load time through performance optimization and drove a 20% increase in user engagement through UX improvements post-launch.

⚙️ SKILLS MATRIX

The Full Tech Stack — By domain, with no filler.

▸ CYBERSECURITY

- Network Security — perimeter defence, traffic analysis, firewall configuration, infrastructure hardening
- Web Application Security — OWASP Top 10 awareness, injection vulnerabilities, authentication flaws
- Digital Forensics — AI-generated image detection, generative artifact analysis, forensic pipeline design
- Vulnerability Assessment — SME security posture evaluation, risk identification and prioritization
- Threat Intelligence — threat actor profiling, phishing vector analysis, incident response principles
- OSINT — open-source intelligence gathering techniques for reconnaissance
- Phishing Detection — recognizing social engineering attacks; email and domain analysis
- Cybersecurity Training — developing and delivering security awareness curriculum to students and SMEs

▸ ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

- Deep Learning — TensorFlow, Keras: model architecture design, training pipelines, evaluation, deployment

- Computer Vision — CNN design, Transfer Learning (Xception, MobileNetV2, EfficientNet), image classification
- Explainable AI — Grad-CAM, activation visualization, model interpretability for forensic and production use
- NLP — tokenization, intent recognition, named entity extraction, text classification, sequence modelling
- LLM Integration — prompt engineering, context window management, multi-turn conversation handling
- Retrieval-Augmented Generation (RAG) — full pipeline: ingest → chunk → embed → retrieve → generate
- Classical ML — Scikit-learn: regression, classification, clustering, cross-validation, feature engineering
- Data Science — Pandas, NumPy, Matplotlib, Seaborn: EDA, data cleaning, statistical analysis, visualization
- HuggingFace — Transformers library, model fine-tuning, inference pipelines, embedding models
- Azure AI — Azure Cognitive Search, Azure OpenAI, cloud-scale RAG architecture (from IEEE paper)

▸ FULL-STACK WEB DEVELOPMENT

- React — component architecture, hooks (useState/useEffect/useContext/useReducer), React Router, state management
- Node.js & Express.js — server architecture, middleware, routing, JWT authentication, REST API lifecycle
- MongoDB — document modelling, aggregation pipeline, indexing strategy, Mongoose ODM
- MySQL — schema design, full normalization (3NF), stored procedures, complex queries, ERD
- RESTful API Design — endpoint architecture, authentication, rate limiting, versioning, error handling
- HTML5 & CSS3 — semantic markup, Flexbox, Grid, animations, pseudo-elements, responsive design
- Bootstrap 5 — grid system, component library, utility-first responsive design
- JavaScript (ES6+) — async/await, Promises, fetch API, DOM manipulation, event-driven architecture

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▸ MOBILE DEVELOPMENT

- Flutter — widget tree architecture, stateful/stateless widgets, navigation, Provider/Bloc state management

- Dart — language fundamentals, async/await, streams, null safety, OOP in mobile context
- Mobile UI/UX — platform design guidelines (Material Design), accessibility, adaptive layouts

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▸ PROGRAMMING LANGUAGES

- Python — primary AI/ML, automation, scripting, web scraping, data engineering, research
- C++ — OOP principles, data structures, algorithms, game logic, system-level application design
- JavaScript (ES6+) — full-stack web: React front-end and Node.js back-end
- SQL — relational design, normalization, complex joins, subqueries, stored procedures
- Dart — Flutter mobile development

▸ TOOLS, DEVOPS & WORKFLOW

- Git & GitHub — version control, branching strategy, pull requests, code review, conflict resolution
- Beautiful Soup & Requests — web scraping, HTML parsing, automated data extraction pipelines
- Schedule / Cron — task automation and pipeline management in Python
- smtplib / Email Automation — notification system engineering in Python
- REST API Consumption — authenticated API integration, OAuth, API key management
- VS Code, PyCharm, IntelliJ — primary development environments
- Postman — API testing and documentation

▸ SOFT SKILLS & PROFESSIONAL

- Technical Writing — IEEE-standard research paper authored and peer-review accepted
- Client Communication — requirements gathering and stakeholder management (Skinsations, ORIP, Sense.ai, Digital Ocean WET)
- Cross-Disciplinary Collaboration — CS × Arts (AfterLight), CS × Research (Grantify), CS × Field Work (WET Program)
- Presentation — Project Display pitches, industry award presentations, hackathon track direction
- Leadership — Head of Neural Nova (BTech Fest), Teaching Assistant, Campus Ambassador, Researcher (CAIR)
- Research Methodology — qualitative interviewing, quantitative survey design, statistical analysis, academic reporting

📬 READY TO COLLABORATE?

Available for internships · Cybersecurity · Software Engineering · AI/ML Research · Full-Stack Development

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